

20879 Project Proposal:

Modeling Historical and Linguistic Relations Between Italian Dialects and Contemporary Languages Through Embedding Spaces*

Group 3: Tommaso Errico, Federico Ferrari, Edoardo Paccagnella,
Raffaele Simbari, Samuele Nicolò Straccialini

April 2026

We will investigate whether word- and sentence-level embedding spaces can recover meaningful similarity patterns that reflect both shared linguistic background and contact-driven influence across language varieties. We focus on Italian dialectal varieties because the Italian dialect landscape is characterized by deep internal variation and a complex history of linguistic contact, making it a particularly compelling domain to investigate. More specifically, we will test whether computational representations are consistent with established hypotheses from historical linguistics and language contact studies, including the expectation that some Italian dialects pattern more closely with neighboring Romance varieties, while others show measurable traces of contact-driven influence from non-Romance languages such as Arabic, Greek, Germanic, or Slavic varieties. We frame the task as a language technology problem of representation learning and similarity modeling.

Using comparable corpora from a selected set of Italian dialects and external language varieties, we construct and compare multiple representations under controlled conditions, including character n-gram TF-IDF baselines, static word embeddings, and sentence-level embedding models, to model similarity at both the lexical and sentence-structural level. We will then compute distances between varieties in these spaces, analyze shifts in shared high-frequency lexical items across varieties, and evaluate the resulting structure through clustering, nearest-neighbor analysis, and low-dimensional visualization.

Our main hypothesis is that embedding-based models capture historically meaningful similarity beyond orthographic or phonetic overlap, and that different representations capture different dimensions of variation, including lexical resemblance, contact-driven borrowing, and sentence-level structure. In particular, we expect surface-based models to be more sensitive to phonetic or orthographic similarity, while more abstract embedding spaces may better reflect patterns of descent, contact, areal proximity, and recurrent structural similarities across comparable sentence types. Evaluation will focus on (i) alignment with accepted linguistic classifications, (ii) the stability and interpretability of the resulting clusters, and (iii) the consistency of lexical and sentence-level neighborhoods across varieties, using metrics such as cosine distance, silhouette score, cluster agreement with external labels, and nearest-neighbor overlap for aligned lexical items and comparable constructions. The project contributes both to computational dialectology and to a broader methodological question in NLP: to what extent can representational similarity serve as evidence of genuine linguistic relatedness in low-resource, highly variable language data?

*Temporary title